

Learning Vector Quantization

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Agenda

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- 2 Architecture
- 3 Algorithm of LVQ
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Learning Vector Quantization

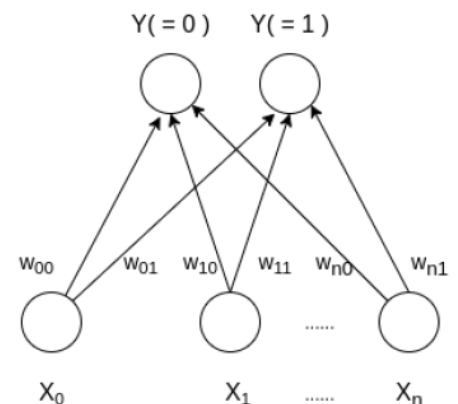
- Learning vector quantization is a competitive network which uses Supervised Learning.
- It is defined as a process of classifying the patterns where each output unit represents a class.
- As it uses supervised learning, the network will be given a set of training patterns with known classifications along with an initial distribution of the output class.
- After completing the training process, LVQ will classify an input vector by assigning it to the same class as that of the output unit.

Architecture of LVQ

- Architecture of LVQ is similar to that of KSOM. The layers are fully connected with having weights on them.

Parameters used :

- X = training vector ($X_1, \dots, X_n$)
- T = class for training vector X
- w_j = weight vector for j th output unit
- c_j = class associated with the j th output unit



LVQ Training Algorithm

Training algorithm:

Step 1: Initialize reference vectors, which can be done as follows -

Step 1a - from the given set of training vectors, take first m number of clusters training vectors and use them as weight vectors. The remaining vectors can be used for training

Step 1b - Assign the initial weight and classification randomly

Step 1c - apply k-means clustering method.

Step 2: initialize reference vector α

Step 3: continue with step 4-9, if the condition stopping this algorithm is not met.

Step 4: Follow step 5-6 for every training i/p vector

Step 5: calculate square of euclidean distance for

$j = 1$ to m and $i = 1$ to n

$$D(j) = \sum_{i=1}^n \sum_{j=1}^m (x_i - w_{ij})^2$$

Step 6: obtain the winning unit J where D_j is minimum.

Step 7: calculate the new weight of the winning unit by the following relation -

if $T = C_j$ then $w_j(\text{new}) = w_j(\text{old}) + \alpha[x - w_j(\text{old})]$

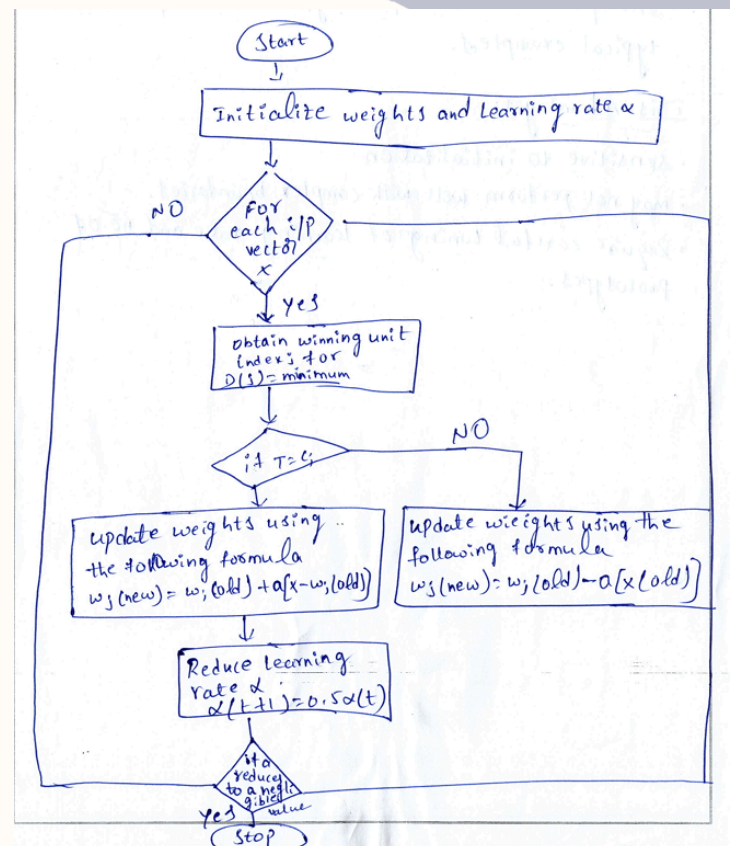
if $T \neq C_j$ then $w_j(\text{new}) = w_j(\text{old}) - \alpha[x - w_j(\text{old})]$

Step 8: Reduce the learning rate α .

Step 9: Test for the stopping condition.

It may be as follows

Flowchart of LVQ



Advantages

Easy to implement and understand

Fast classification after training

Interpretable model-prototype vectors
represents typical examples

Disadvantages

Sensitive to initialization

May not perform well with
complex boundaries

Require careful tuning of learning rate
and no.of prototypes

Thank you!